Linear Algebra Review

ECE/MAE 5930 Mobile Robotics

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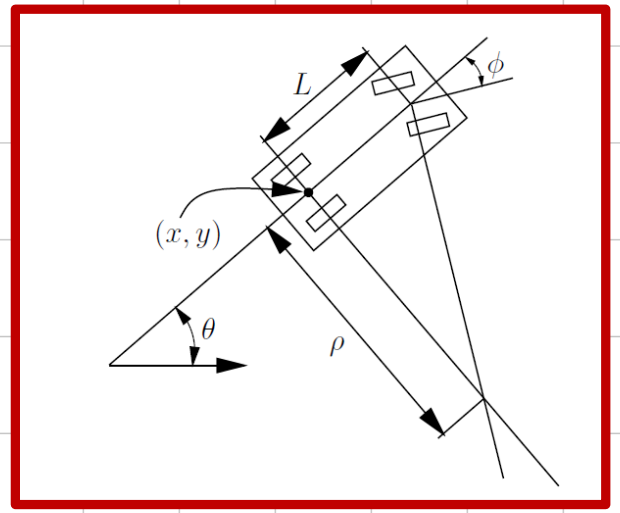
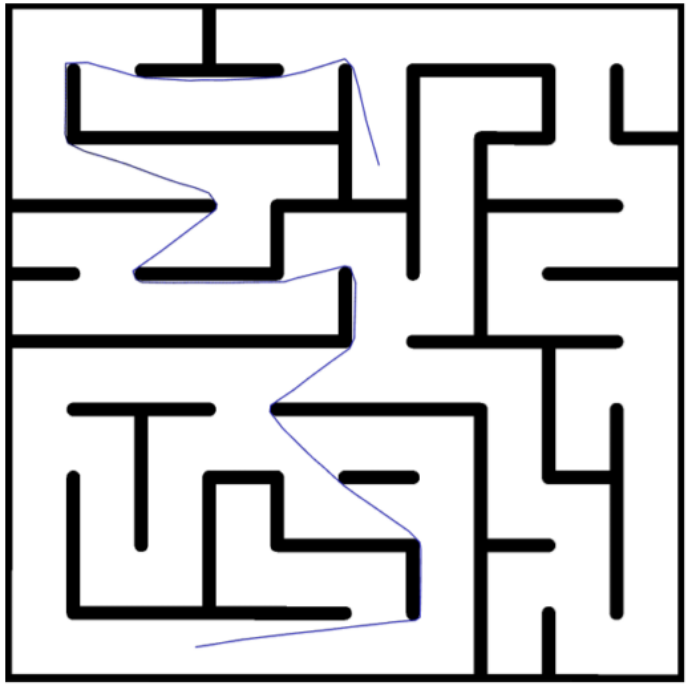
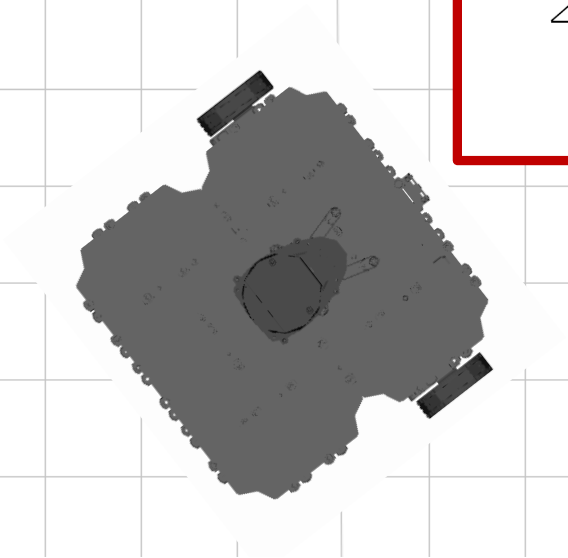
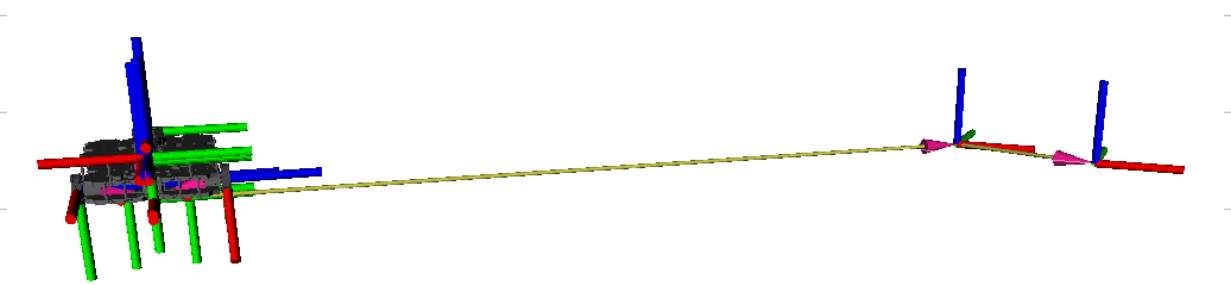
Motivation

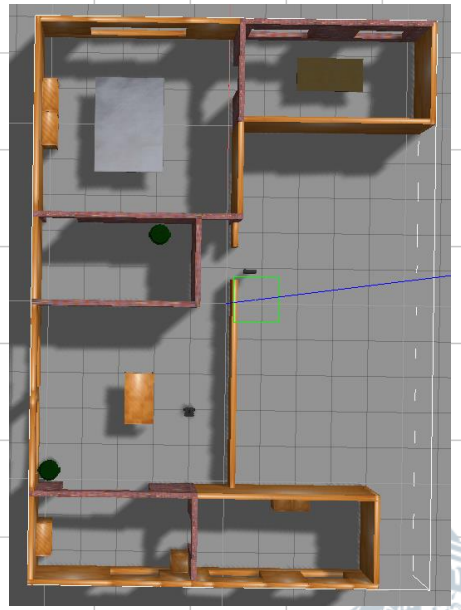
- Reference frames
- Obstacle maps
- Vehicle motion











Overview

- Set notation
  - Set operations: union, intersections, differences
- Real vector space
  - Idea
  - Linear independence
  - Dimension
  - Products: dot, cross
  - Euclidean space
- Mappings
  - General idea
  - Linear mapping
  - Special mapping: orthogonal matrices
- Groups

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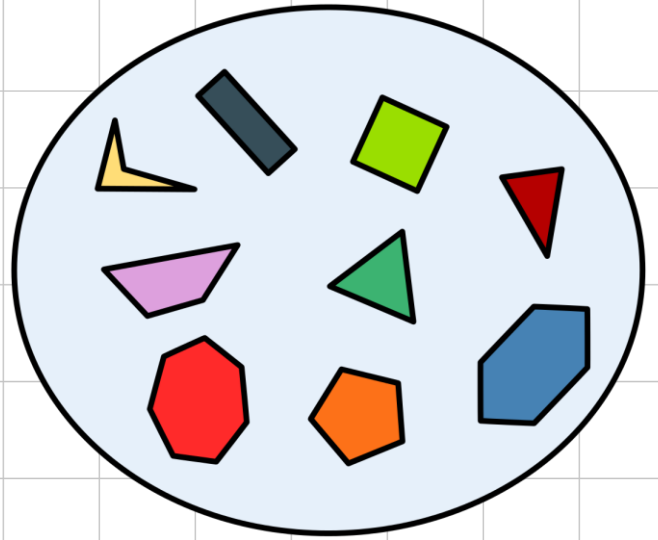
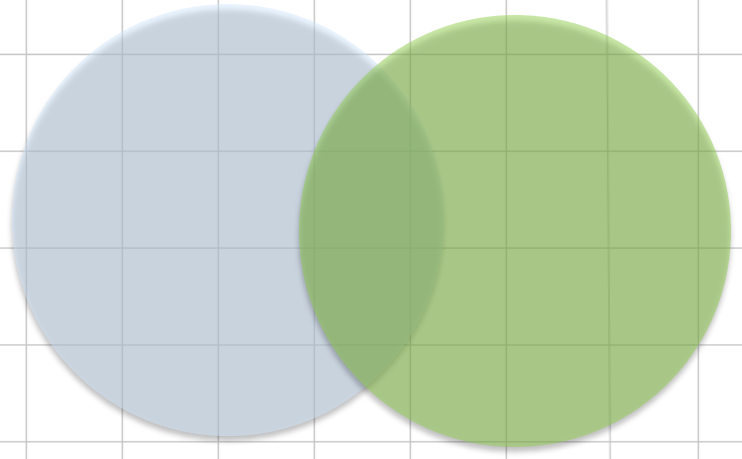
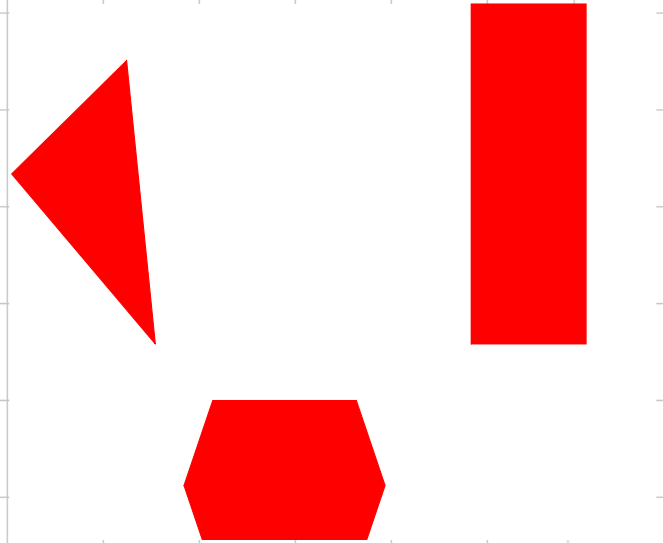
Set Notation

- Set

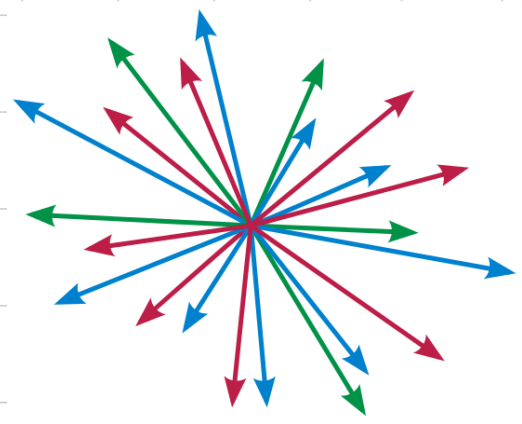


- Basic operations (union, intersection, difference)





Vector Space

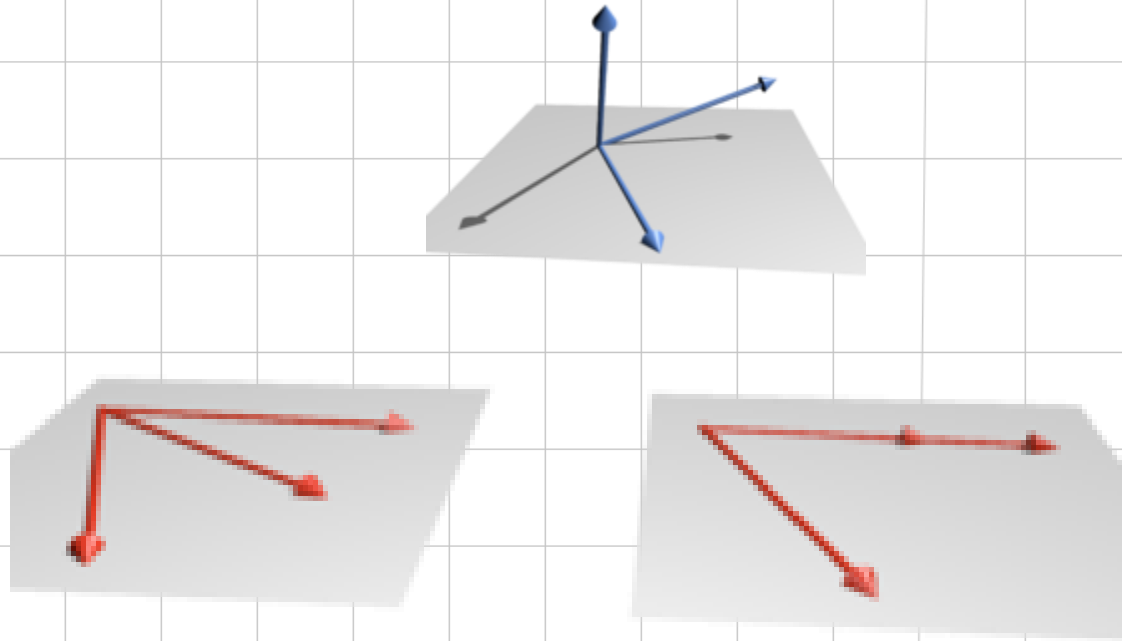


- Definition

- Dimension

Linear Independence

- Idea



- Span

Dot Product

- Dot product

- Properties

Cross Product

- Definition for 3 dimensions

- Properties

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- Defining right-hand basis vectors

Metric Space

- Idea: Associate coordinates with a metric (i.e. a ruler)

- Example: Euclidean space
  - Metric: dot product

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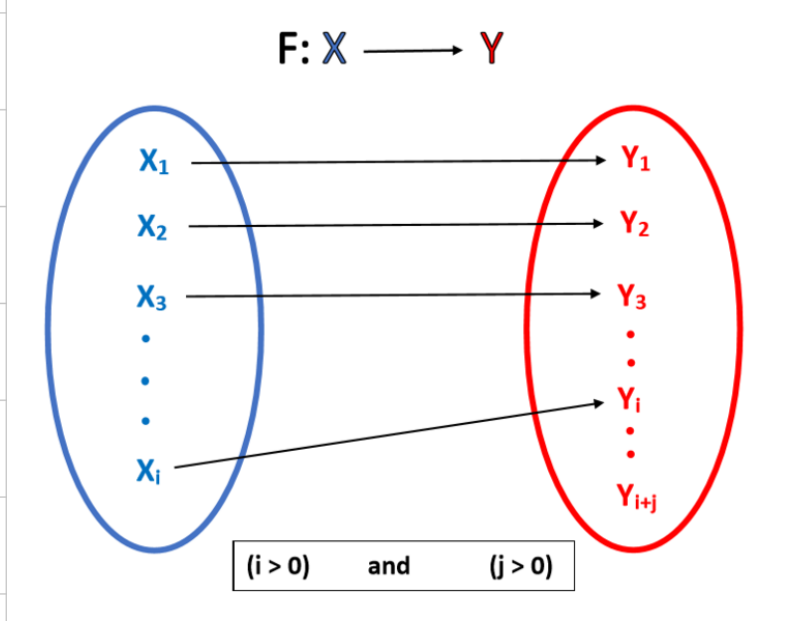
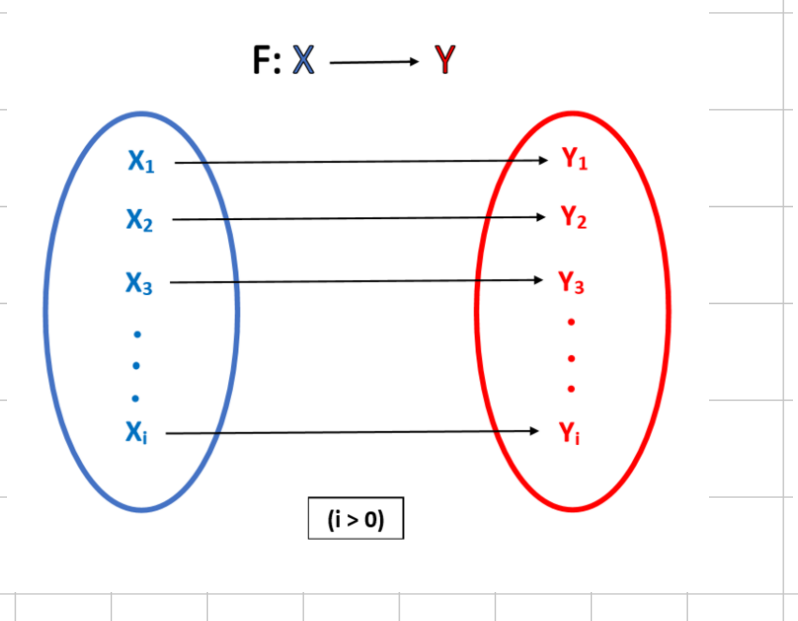
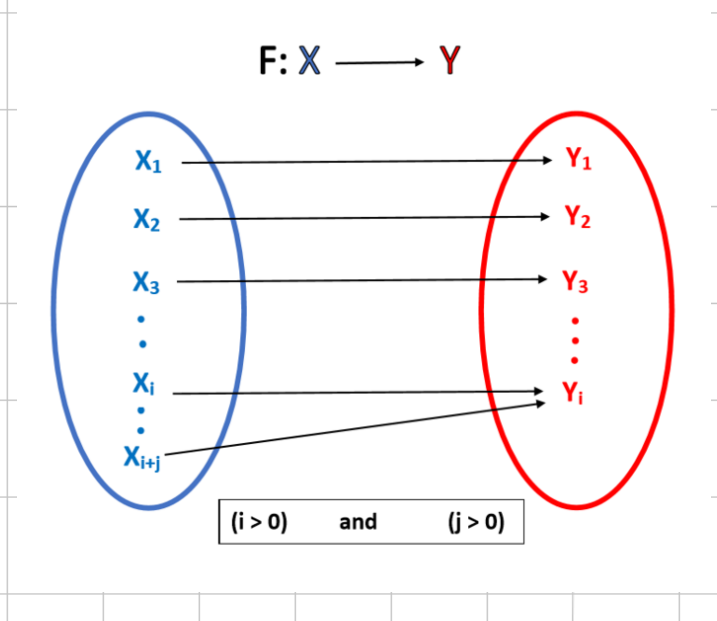
Mapping

- Idea: map from one set to another set







Linear Mapping

- Definition
- Matrix mapping:

Special Mapping: Orthogonal Mapping

- Orthogonal mapping / transformation:

$a' = Oa \qquad b' = Ob$

$\langle a', b' \rangle = (Oa)^T Ob = a^T \cancelto{I}{O^T O} b = a^T b$

for orthogonal matrix

$O^T = O^{-1}$

- Special orthogonal group (rotation matrix, direction cosine matrix)

$SO(n) : O \in SO(n) \Rightarrow \det(O) = 1 \qquad O^{-1} = O^T$

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Group

- A group is a set with the following properties
  - Closure under group operator $g_1 \in G$ & $g_2 \in G \Rightarrow g_1 \circ g_2 \in G$
  - An identity element $g_1 \in G$ $g \circ I = g$ $I \in G$
  - Inverse element $g_1 \in G \Rightarrow \exists\ g_2 \in G$ s.t. $g_1 \circ g_2 = I$
  - Associativity $g_1 \circ (g_2 \circ g_3) = (g_1 \circ g_2) \circ g_3$
- Back to the special orthogonal group

Special orthogonal

Matrix Multiplication

Identity Matrix

Transpose (Inverse)

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Summary

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